S5 RQ1: Full Metric Breakdown

This section provides supporting analyses for RQ1. Table S5.1 compares the seed and best-pairwise harness across all metrics. Table S5.2 reports how pairwise accuracy correlates with other metrics, and Table S5.3 counts how many harnesses improve over the seed. Overall, pairwise accuracy aligns with accuracy and precision, but not consistently with recall, CWE identification, localization, or Joint.

Table S5.1: Seed and best-pairwise harness performance on TitanVul+CWEval (202 pairs). GLM-5.3 ties at iterations 4 and 6.

Agent + Model	Harness	Pairwise	Accuracy	Precision	Recall	F1	CWE acc	Loc F1	Loc hit	Joint
Codex + Inkling-Small	Seed	6.4	0.520	0.511	0.955	0.665	30.8	33.0	61.7	15.0
	Best pairwise (iter 12)	**27.7**	0.589	0.572	0.695	0.627	18.8	28.5	59.7	9.4
Codex + DeepSeek V4 Flash	Seed	14.4	0.552	0.531	0.896	0.667	42.1	32.0	69.1	22.8
	Best pairwise (iter 11)	**36.1**	0.651	0.660	0.624	0.641	36.0	28.6	51.6	20.9
Claude Code + Opus 4.6	Seed	10.4	0.542	0.524	0.926	0.669	46.2	36.4	68.5	24.4
	Best pairwise (iter 3)	**33.7**	0.653	0.634	0.728	0.677	39.8	39.8	67.3	28.6
Claude Code + GLM-5.3	Seed	19.8	0.572	0.549	0.807	0.653	40.0	32.3	73.0	21.9
	Best pairwise (iter 4)	**31.2**	0.616	0.609	0.648	0.628	40.4	32.6	72.5	29.8
	Best pairwise (iter 6)	**31.2**	0.635	0.641	0.615	0.628	47.7	37.4	71.5	34.1

Table S5.2: Correlation between pairwise accuracy and other metrics. Positive values mean the metrics improve together; negative values indicate a trade-off.

Metric	Inkling-Small	DeepSeek V4 Flash	Opus 4.6	GLM-5.3
Accuracy	$+0.88$	$+0.86$	$+1.00$	$+0.64$
Precision	$+0.92$	$+0.89$	$+0.40$	$+0.76$
Recall	-0.72	-0.21	0.00	-0.77
F1	-0.61	$+0.07$	$+0.40$	-0.47
CWE acc (any-match)	-0.63	-0.33	-0.50	$+0.33$
CWE acc (first CWE)	-0.54	-0.28	-0.10	$+0.80$
Loc precision	-0.36	$+0.19$	$+0.40$	$+0.08$
Loc recall	$+0.02$	-0.19	-0.90	-0.04
Loc F1	-0.26	-0.27	$+0.10$	$+0.29$
Loc hit rate	$+0.20$	-0.43	-0.60	$+0.02$
Joint (any-match CWE)	-0.73	-0.56	-0.80	$+0.44$
Joint	-0.64	-0.54	-0.10	$+0.84$

Table S5.3: Number of harnesses that beat the seed on pairwise accuracy, localization, and Joint. The last two columns show the largest gains over the seed.

	Harnesses beating the seed on				Largest gain over seed	
Agent + Model	Pairwise	+ Loc hit	+ Loc F1	+ Joint	Pairwise	Loc F1
Codex + Inkling-Small	15 / 15	9	2	2	$+21.3$	$+0.9$
Codex + DeepSeek V4 Flash	13 / 13	0	1	4	$+21.8$	$+3.7$
Claude Code + Opus 4.6	4 / 4	1	4	3	$+23.3$	$+4.4$
Claude Code + GLM-5.3	8 / 8	0	6	7	$+11.4$	$+5.1$
Total	40 / 40	10	13	16		